

Trigonometry MCQ Step-by-Step Solutions

File Name: 31.jpg

Question: Find the value of $2 \tan 60^\circ / (1 - \tan^2 60^\circ)$.

Step 1: Substitute the standard value $\tan 60^\circ = \sqrt{3}$.

Step 2: Expression = $2(\sqrt{3}) / (1 - (\sqrt{3})^2) = 2\sqrt{3} / (1 - 3)$.

Step 3: Simplification = $2\sqrt{3} / (-2) = -\sqrt{3}$.

Correct Option: (D)

File Name: 32.jpg

Question: If $\operatorname{cosec} B = 2/\sqrt{3}$ and $A + B = 90^\circ$, find the value of $\sec A$.

Step 1: Since $\operatorname{cosec} B = 2/\sqrt{3}$, the standard angle is $B = 60^\circ$.

Step 2: Use $A + B = 90^\circ$ to find A: $A = 90^\circ - 60^\circ = 30^\circ$.

Step 3: Calculate $\sec A = \sec 30^\circ = 2/\sqrt{3}$.

Correct Option: (A)

File Name: 33.jpg

Question: Find the maximum value of $1 / \operatorname{cosec} \theta$, for $(0^\circ \leq \theta \leq 90^\circ)$.

Step 1: Simplify the identity: $1 / \operatorname{cosec} \theta = \sin \theta$.

Step 2: In the given domain $0^\circ \leq \theta \leq 90^\circ$, the value of $\sin \theta$ ranges from 0 to 1.

Step 3: The maximum value of $\sin \theta$ occurs at $\theta = 90^\circ$, which is 1.

Correct Option: (B)

File Name: 34.jpg

Question: The value of $(1 - \tan^2 45^\circ) / (1 + \tan^2 45^\circ)$ is equal to...

Step 1: Substitute the standard value $\tan 45^\circ = 1$.

Step 2: Expression = $(1 - 1^2) / (1 + 1^2) = (1 - 1) / (1 + 1) = 0 / 2 = 0$.

Step 3: Match with options: $\tan 0^\circ = 0$.

Correct Option: (D)

File Name: 35.jpg

Question: If $x \tan 45^\circ \cos 60^\circ = \sqrt{3} \sin 60^\circ \cot 60^\circ$, find x.

Step 1: Substitute values: $\tan 45^\circ = 1$, $\cos 60^\circ = 1/2$, $\sin 60^\circ = \sqrt{3}/2$, $\cot 60^\circ = 1/\sqrt{3}$.

Step 2: $\text{LHS} = x * 1 * (1/2) = x / 2$.

Step 3: $\text{RHS} = \sqrt{3} * (\sqrt{3}/2) * (1/3) = (3/2) * (1/\sqrt{3}) = \sqrt{3}/2$.

Step 4: Equating both sides: $x / 2 = \sqrt{3} / 2 \Rightarrow x = \sqrt{3}$.

Correct Option: (C)

File Name: 36.jpg

Question: Find the value of $\cos 60^\circ - \operatorname{cosec} 30^\circ + \tan 45^\circ$.

Step 1: Substitute standard values: $\cos 60^\circ = 1/2$, $\operatorname{cosec} 30^\circ = 2$, $\tan 45^\circ = 1$.

Step 2: Expression = $(1/2) - 2 + 1 = (1/2) - 1 = -1/2$.

Correct Option: (B)

File Name: 37.jpg

Question: If $2 \cos \theta = 1$, then find the value of θ .

Step 1: Isolate the cosine function: $\cos \theta = 1/2$.

Step 2: Identify the corresponding principal angle: $\theta = 60^\circ$.

Correct Option: (b)

File Name: 38.jpg

Question: If $\sin A = 3/5$, then find the value of $\cot A$.

Step 1: Let opposite side = 3 and hypotenuse = 5.

Step 2: Find adjacent side via Pythagoras theorem: $\sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4$.

Step 3: Compute $\cot A = \text{adjacent} / \text{opposite} = 4 / 3$.

Correct Option: (B)

File Name: 39.jpg

Question: The expression $2 \tan 30^\circ / (1 - \tan^2 30^\circ)$ is equal to...

Step 1: Use the double-angle identity: $\tan 2\theta = 2 \tan \theta / (1 - \tan^2 \theta)$.

Step 2: Here, $\theta = 30^\circ$, so the expression becomes $\tan(2 * 30^\circ) = \tan 60^\circ$.

Correct Option: (C)

File Name: 40.jpg

Question: If $\cos A = 1/2$, then $\tan A$ is equal to...

Step 1: Since $\cos A = 1/2$, the standard angle is $A = 60^\circ$.

Step 2: Calculate $\tan A = \tan 60^\circ = \sqrt{3}$.

Correct Option: (B)